

Idea Submission Template for AI Unlocked

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Challenge Area:	Track 2 - AI for India	Registered Email ID of Team Lead:	22je0381@iitism.ac.in

Project Title

VyaparSahayak

Project Sub-title: Voice-First, Offline-First Inventory & Credit Assistant for India's 100 Million Weekly Bazaar Vendors

The Team

I am **Harsh Porwal**, a **solo developer** and a 3rd year student (graduating 2027) in **Mathematics and Computing** at **IIT (ISM) Dhanbad**, also serving as an **AI/ML Intern at DGIS (Indian Army)**.

My inspiration is deeply personal: **my father is a weekly bazaar vendor in Delhi**. I have grown up watching him — and thousands like him — struggle with 'mental math' for inventory, losing money on unsold perishable stock, and forgetting *udhaar* (credit) details because existing apps are too complex and text-heavy.

I decided to participate in AI Unlocked alone because I have a singular, clear vision: to build the tool I wish my father had — a simple, voice-based AI that acts as a **'Munim-ji' (Accountant)** for the illiterate or semi-literate vendor.

The Concept

VyaparSahayak is a **Voice-First, Vernacular AI assistant** designed specifically for street vendors and micro-merchants who operate in chaotic, low-internet environments. It solves three critical problems:

- * **Inventory Forecasting:** A time-series model (ARIMA/LSTM) predicts demand — e.g., "Buy 20 kg less onions tomorrow, prices are dropping."
- * **Voice Ledger:** Instead of typing, the vendor simply speaks: "Raju took 500 rupees credit." The AI parses the dialect and updates the digital ledger.
- * **Offline-First Architecture:** Works fully offline and syncs data when the vendor reaches a Wi-Fi zone or stable network — crucial for spotty 4G areas.

Unlike generic fintech apps, VyaparSahayak is built for the *informal economy*, prioritising voice interaction in local dialects (Hindi, Bhojpuri, Haryanvi) over text menus.

Target Audience or Market

Primary: 50-100 million micro-vendors in India (weekly bazaar sellers, *thela* owners, street hawkers).

Geography: Tier-2 and Tier-3 cities, and urban slums in metros like Delhi and Mumbai.

Demographics: Low-literacy, daily-wage earners who own smartphones but struggle with complex UIs.

Market Size: Government data identifies 49.48 lakh (approx. 4.9 million) registered street vendors (The Tribune, Government of India). Informal economy estimates reach 10 million+ vendors nationwide — a massive, underserved segment of India's informal economy with no adequate digital tools today.

Personas

Persona 1 — Rajesh, 45, Vegetable Vendor (Delhi Weekly Bazaar)

Uses WhatsApp but finds typing difficult. Pain point: regularly overbuys perishable stock (tomatoes, greens) which rots if unsold — causing ~20% losses. Keeps credit records in a paper notebook that gets wet or lost. **Goal:** know exactly how much to buy for tomorrow's market and track who owes him money — without writing anything down.

Persona 2 — Sita, 32, Kirana Store Owner (Tier-2 Town)

Small shop owner who wants simple, reliable reorder suggestions and a visible digital ledger she can present to local microcredit programs for working-capital loans.

End-to-End Sequence Diagram

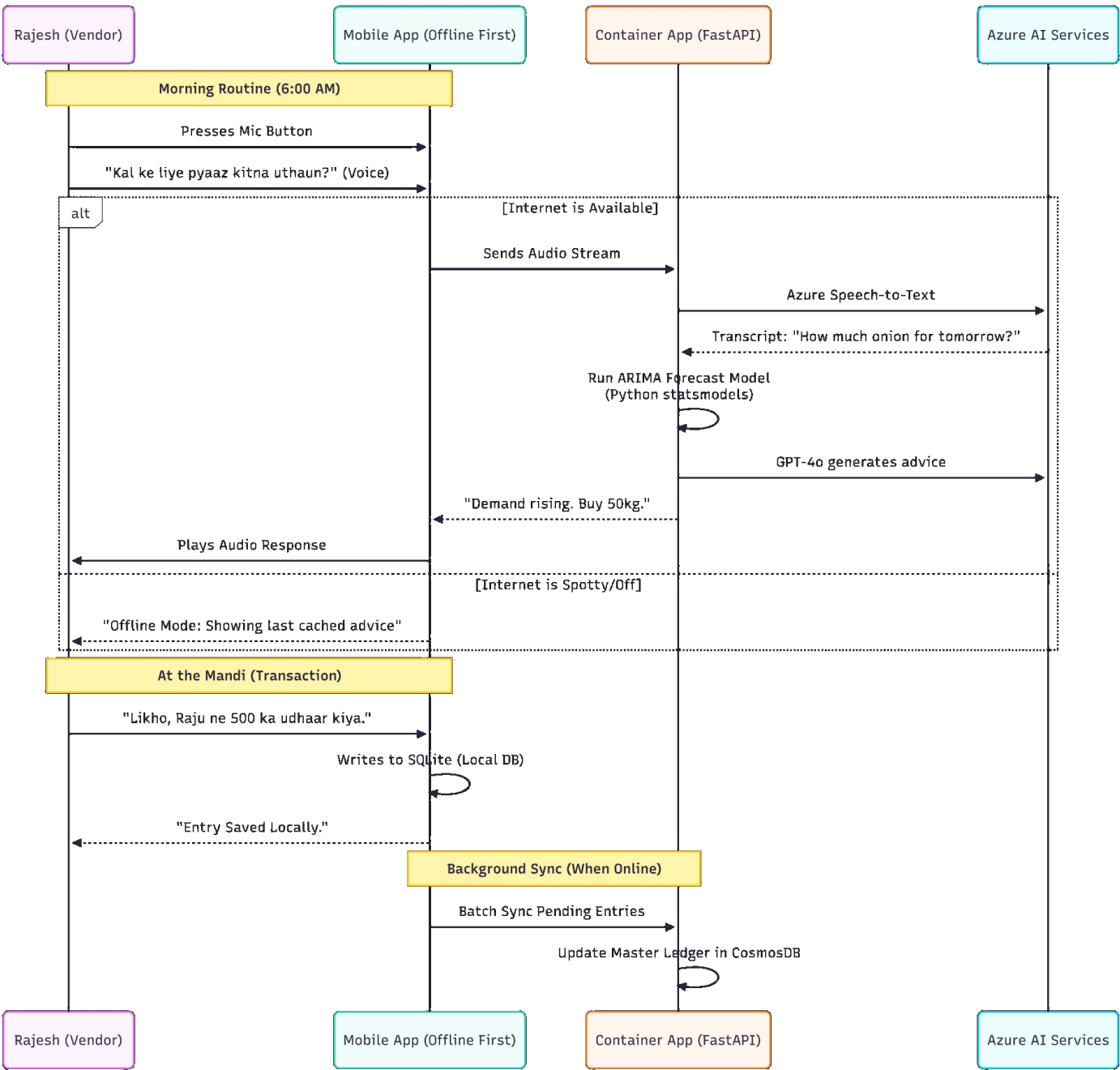


Fig. 3: Sequence Diagram — Rajesh (Vendor) morning routine: voice query -> Azure Speech-to-Text -> ARIMA forecast -> GPT-4o advice -> audio response. Offline mode: cached advice. Mandi transaction: SQLite write -> background sync to CosmosDB.

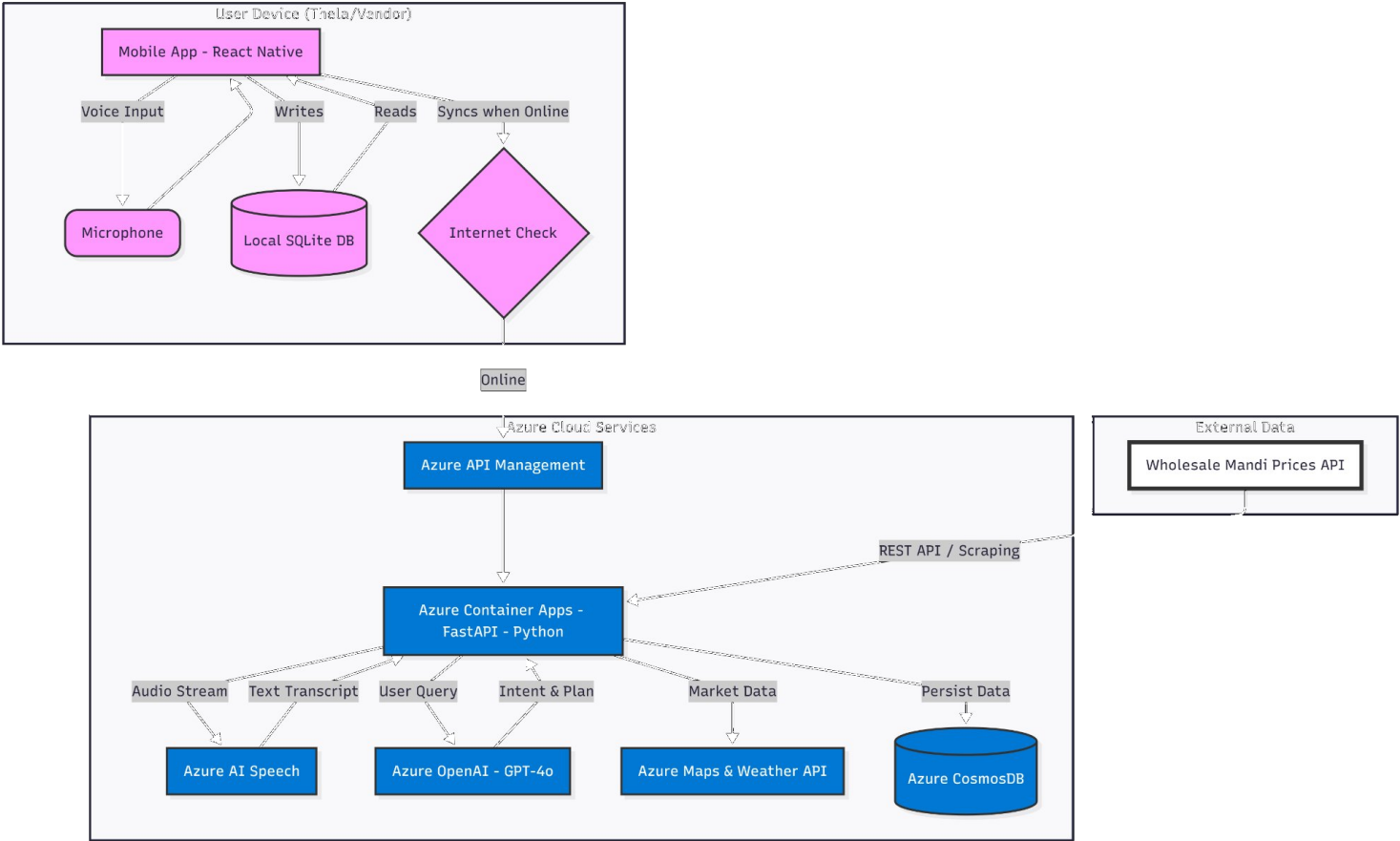


Fig. 1: VyaparSahayak Architecture — User Device (React Native + SQLite) -> Internet Check -> Azure API Management -> Azure Container Apps (FastAPI/Python) -> Azure AI Speech | Azure OpenAI GPT-4o | Azure Maps | Azure CosmosDB; External: Wholesale Mandi Prices API

How It Works

VyaparSahayak uses a 'Listen - Process - Act' loop powered by three cooperating agents:

- * **Planner (Azure OpenAI GPT-4o):** Converts spoken inventory or udhaar entries into a structured weekly operational plan — reorder timing, cash target, reminders.
- * **Retriever:** Fetches local mandi prices, short-term weather (Azure Maps), and holiday calendars with provenance/confidence scores. Pulls wholesale prices via REST API / scraping.
- * **Executor:** Runs a lightweight 7-day ARIMA demand forecast (Python statsmodels) and computes reorder quantities plus udhaar credit-risk reminders.

Offline-first flow: The React Native app writes all transactions to a Local SQLite DB instantly. An Internet Check routes live requests to Azure Cloud Services or serves cached advice when offline. On reconnect, pending entries batch-sync to Azure CosmosDB.

Core Technologies

Technology	Role
Azure AI Speech (STT/TTS)	Accurate speech recognition & synthesis in Indian dialects (Hindi, Bhojpuri, Haryanvi)
Azure OpenAI Service (GPT-4o)	Intent extraction, NLU, weekly plan generation, dialect-aware advice
Azure Maps & Weather API	Hyper-local weather & event data for demand prediction
Azure Container Apps + ACR	Scalable FastAPI (Python) microservice deployment
Azure API Management	Secure, rate-limited API gateway for the mobile app
Azure CosmosDB	Master ledger & historical data store; offline-sync target
React Native (Mobile App)	Cross-platform offline-first app with Local SQLite DB
Python / statsmodels (ARIMA)	Lightweight 7-day demand forecasting on the container
Wholesale Mandi Prices API	External data source for real-time wholesale price signals (REST / scraping)

The Business Plan

- * **Freemium Onboarding:** Core 'Voice Ledger' is free — zero barrier to entry for low-income vendors to build the user base.
- * **Premium Insights (Rs. 49/month):** Advanced inventory prediction, market price alerts (wholesale mandi rate scraping), and smart udhaar reminders via SMS.
- * **NGO / Government Pilots:** Free pilot deployments to build trust and data, followed by NGO dashboard subscriptions for KPI tracking (stockouts avoided, time saved).
- * **B2B Data Play:** Aggregated, anonymised data on informal-economy price trends sold to FMCG companies and government planning agencies.
- * **Long-Run — Micro-Finance Referrals:** Embed optional micro-savings or microcredit referral partnerships (compliance & KYC required before integration).

